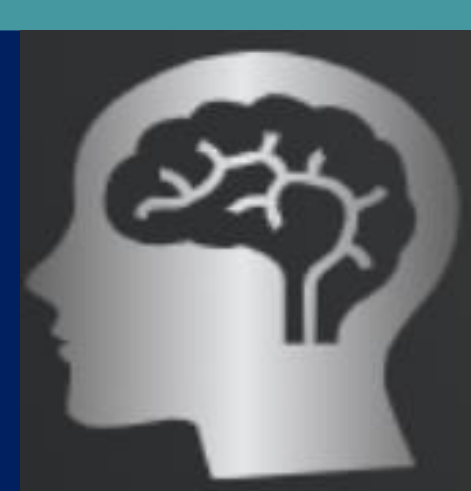




NeuroAID: An Explainable Artificial Intelligence (XAI) Based Brain Tumor Detection System Using MRI Analysis



INTRODUCTION

- Brain tumors are associated with the abnormal growth of brain cells, forming a mass, putting pressure on healthy parts of the brain and spreading there. Globally, 400,000 cases reported in 2020 with a high mortality rate of 60%.
- Detecting tumors in their early stages is crucial for devising personalized treatment and improving chances of survival because average doubling time for gliomas is 14-21 days.
- Magnetic Resonance Imaging (MRI), with gadolinium dye as a visual marker, is used for screening brain tumors due to its non-invasive aspect, high resolution and detail.

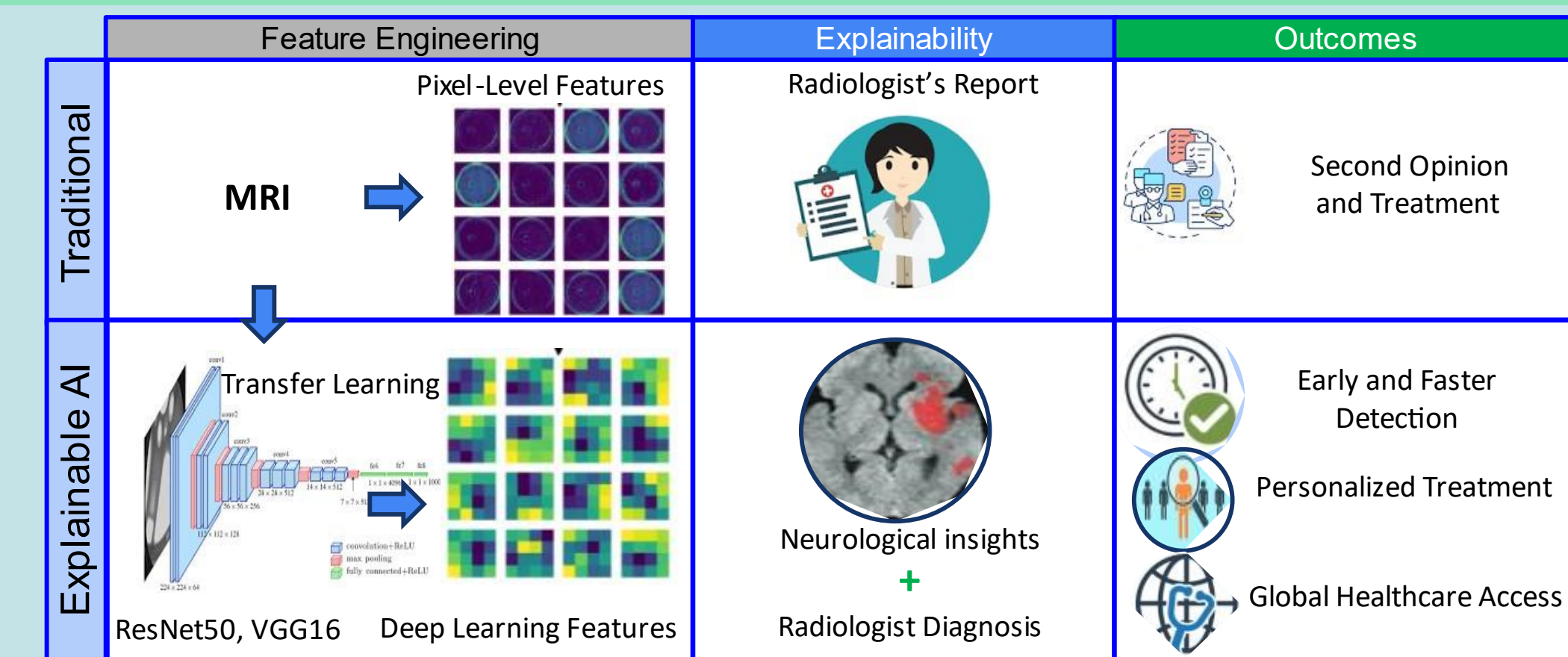
Key Challenges

- Shortage of radiologists in underdeveloped regions: 1/1000
- Error prone: Misdiagnosis in early stages for 4% of patients
- Significant delay in detection (average of 153 days delay)
- Expensive & time consuming (results take up to 2 weeks)
- Confounding with other diseases

XAI-Enabled Rapid Brain Tumor Detection

- Machine Learning (ML) / Deep Learning (DL) techniques trained on hidden features act as **Digital Radiologist**.
- Convolution Neural Network (CNN) classification model is built with many layers of neurons (combination of variables & image features).
- Transfer learning (TL): A CNN model (VGG16) previously trained on a huge dataset is enhanced to achieve robust prediction performance.

Explainability: Provides medical insights and interpretable diagnosis at (a) cohort level (b) patient level to build confidence in personalized treatments.



RESEARCH STUDY

Goal: Develop a novel, explainable algorithm for automated, rapid and robust detection/classification of brain tumors.

RESEARCH QUESTION

Can an XAI hybrid model outperform existing AI methods in detecting brain tumors (glioma, meningioma, & pituitary) and provide neurological insights?

HYPOTHESIS

If the training data set is consistent, balanced and has sufficient diversity of features representing all the classes, then the transfer-learning based hybrid models can classify the tumors & provide explainable neurological insights.

DESIGN SPACE

Independent Variables:

1. AI Algorithms

AI Algorithms	Detection	Classification
ML		
K-neighbors Classifier (KNC)	✓	✓
Naive Bayes (NB)	✓	✓
Logistic Regression (LR)	✓	✓
XGBoost (XGB)	✓	✓
Random Forest (RF)	✓	✓
DL		
Multi-Layer Perceptron (MLP)	✓	✓
CNN	✓	✓
Transfer Learning		
VGG16	✓	✓
Feature-based ML		
VGG16+KNC (KNCV)	✓	✓
ResNet50+KNC (KNCR)	✓	✓
Feature Fusion-based ML		
VGG16+ResNet50 + KNC (KNCF)	✓	✓

2. Algorithmic Parameters

AI Algorithms	Hyperparameters	Range/Settings
ML		
K-neighbors Classifier (KNC)	n-neighbors	[1,50]
XGBoost (XGB)	Learning Rate	[0.01,1]
Random Forest (RF)	Number of Estimators	[10,200]
DL & Transfer Learning		
Multi-Layer Perceptron (MLP)	Number of Layers	[1-5]
	Neurons per layer	[10,100,500]
CNN, VGG16, KNCF, KNCV, KNCR	Optimizer	Adam, SGD
	Batch Size	Various
	Epochs	[1,50]

Dependent Variables:

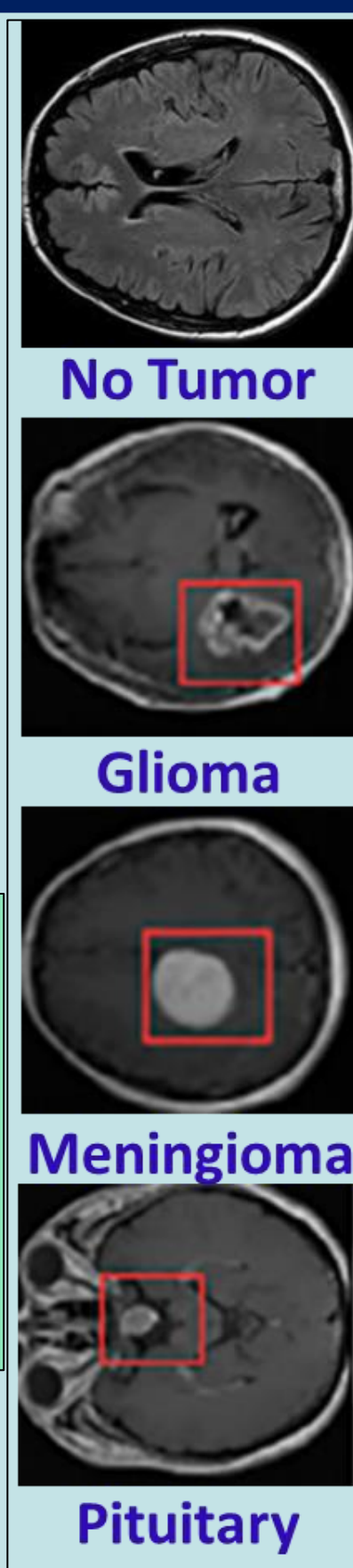
- Accuracy, $A = \frac{TP+TN}{TP+FP+TN+FN}$
- Precision, $P = \frac{TP}{TP+FP}$
- Recall, $R = \frac{TP}{TP+FN}$
- F1 Score = $\frac{2 \cdot P \cdot R}{(P+R)}$

	PREDICTED	
	No Tumor	Tumor
ACTUAL	No Tumor	True Negative (TN) Type II Error
	Tumor	False Negative (FN) Type I Error
		True Positive (TP)

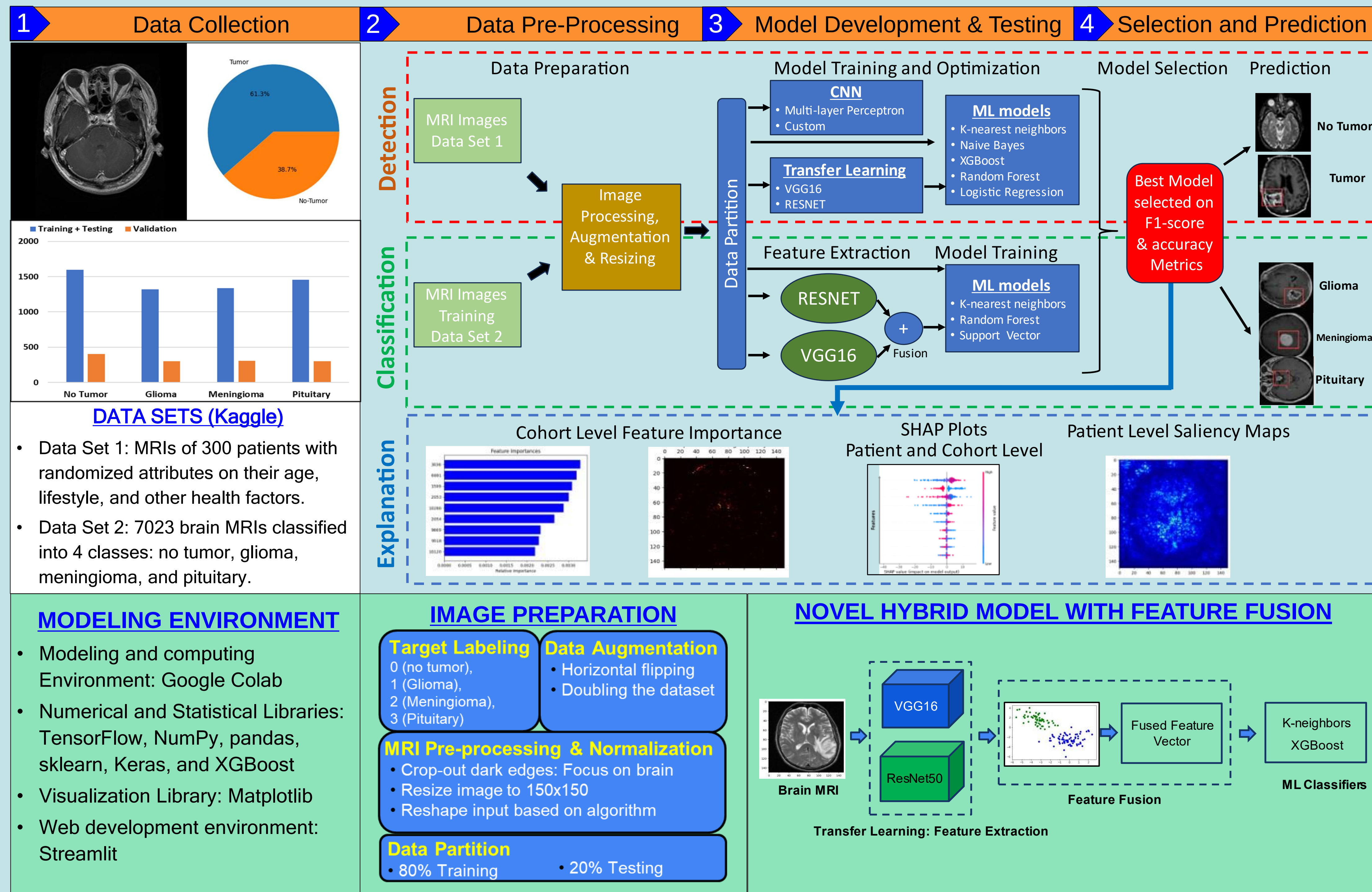
Control: Validation dataset

Controlled Variables:

- Preprocessing & size of MRIs
- Data partitions



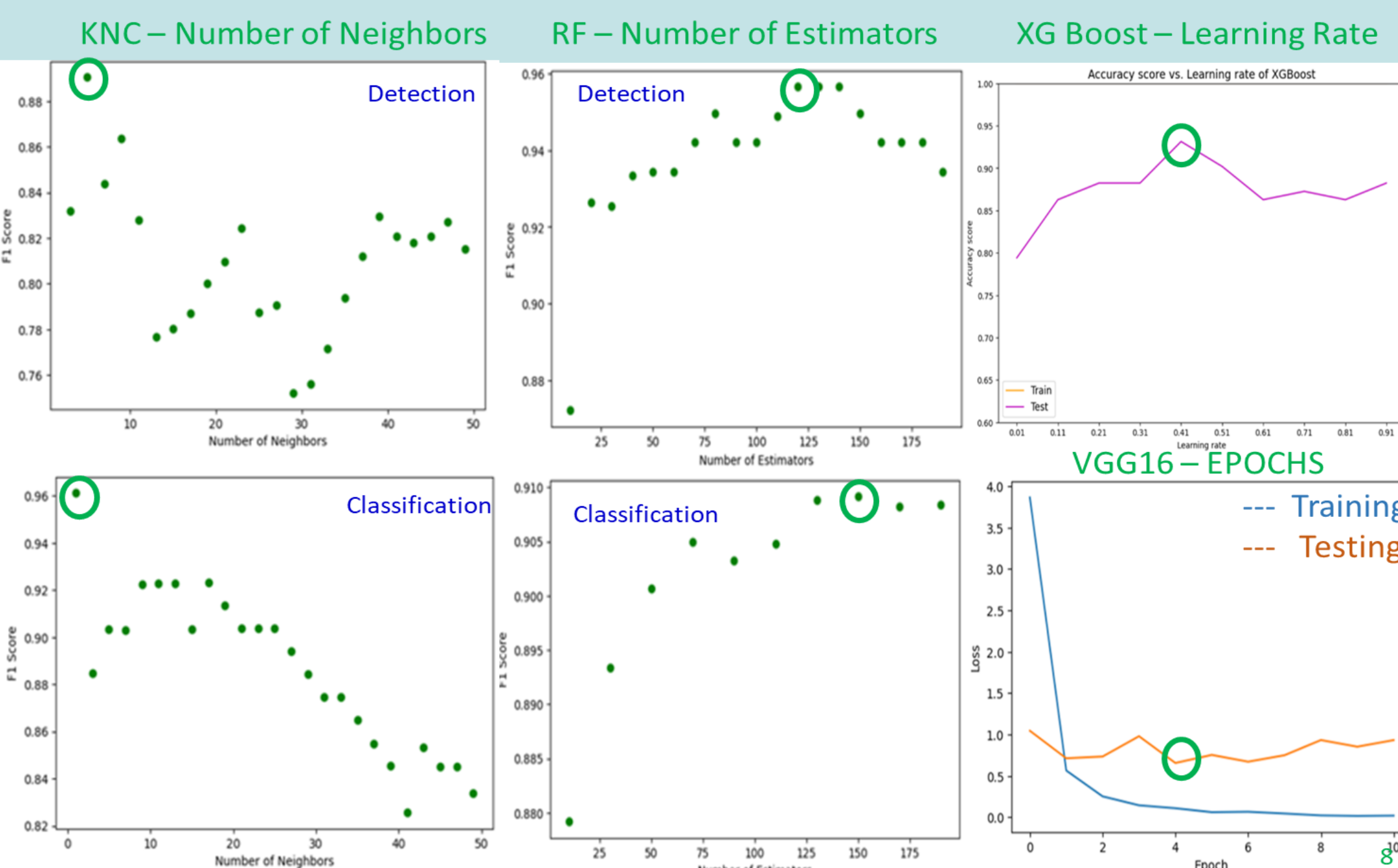
METHODOLOGY



RESULTS

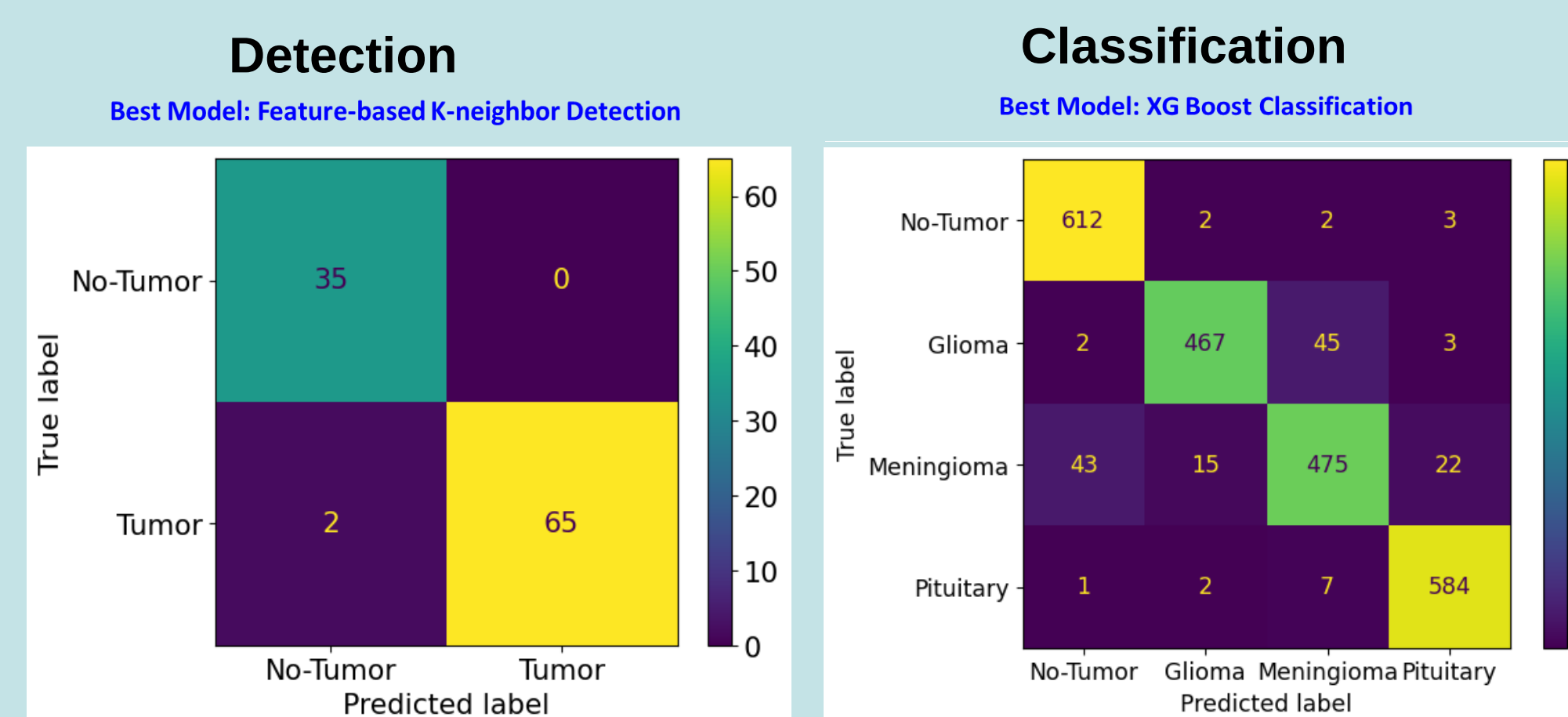
MODEL TRAINING

Hyperparameter Optimization: Parametric study for performance parameters



The green circles indicate the optimal parameters for different algorithms maximizing the F1 score: 5 neighbors in the K-neighbors model for detection and at learning rate of 0.41 for XGBoost algorithm for classification.

CONFUSION MATRICES



MODEL PERFORMANCE METRICS

Detection Models

ML/DL Detection Algorithms	Input Image size	Accuracy	Precision	Recall	F1 Score
K-neighbors Classifier (KNC), n = 5	(*, 22500)	0.86	0.93	0.85	0.89
Naive Bayes (NB)	(*, 22500)	0.69	0.89	0.6	0.71
Logistic Regression (LR)	(*, 22500)	0.83	0.86	0.9	0.88
XGBoost (XGB), lr=0.41	(*, 22500)	0.93	0.92	0.99	0.95
Random Forest (RF)	(*, 22500)	0.94	0.93	0.99	0.96
Convolutional Neural Network(CNN)	(*,150,150,1)	0.90	0.91	0.96	0.93
Transfer Learning (VGG16)	(*,150,150,3)	0.94	0.96	0.96	0.96
VGG16 Feature + KNC (KNCV)	(*,150,150,3)	0.96	1	0.94	0.97
ResNet50 Feature + KNC (KNCR)	(*,150,150,3)	0.98	1	0.97	0.98
Fused (VGG16+ResNet50) + KNC (KNCF)	(*,150,150,3)	0.97	1	0.96	0.98

Classification Models

ML/DL Classification Algorithms	Input Image size	Accuracy	Precision			Recall			F1-Score		
			G	M	P	G	M	P	G	M	P
Best K-neighbors Classifier (KNC), n = 1	(*, 22500)	0.93	0.87	0.92	0.96	0.91	0.86	0.98	0.89	0.89	0.97
Naive Bayes (NB)	(*, 22500)	0.62	0.5	0.39	0.79	0.83	0.33	0.63	0.62	0.36	0.7
XGBoost (XGB)	(*, 22500)	0.94	0.97	0.9	0.97	0.9	0.97	0.98	0.93	0.88	0.97
Random Forest (RF)	(*, 22500)	0.91	0.94	0.84	0.92	0.84	0.83	0.97	0.89	0.83	0.95
Transfer Learning (VGG16)	(*,150,150,3)	0.93	0.93	0.82	0.96	0.89	0.93	0.93	0.91	0.87	0.95
VGG16-Feature + KNC (KNCV)	(*,150,150,3)	0.93	0.91	0.88	0.93	0.88	0.86	0.98	0.89	0.87	0.95
ResNet50-Feature + KNC (KNCR)	(*,150,150,3)	0.93	0.87	0.89	0.86	0.86	0.86	0.99	0.87	0.88	0.97
Fused(VGG16+ResNet50) + KNC (KNCF)	(*,150,150,3)	0.93	0.91	0.88	0.93	0.88	0.86	0.98	0.89	0.87	0.95

EXPLAINABILITY

Cohort Level

Feature Importance:

Top 10 important pixels for the overall classification.

Heat Maps:

Important pixels on the standardized image shape.

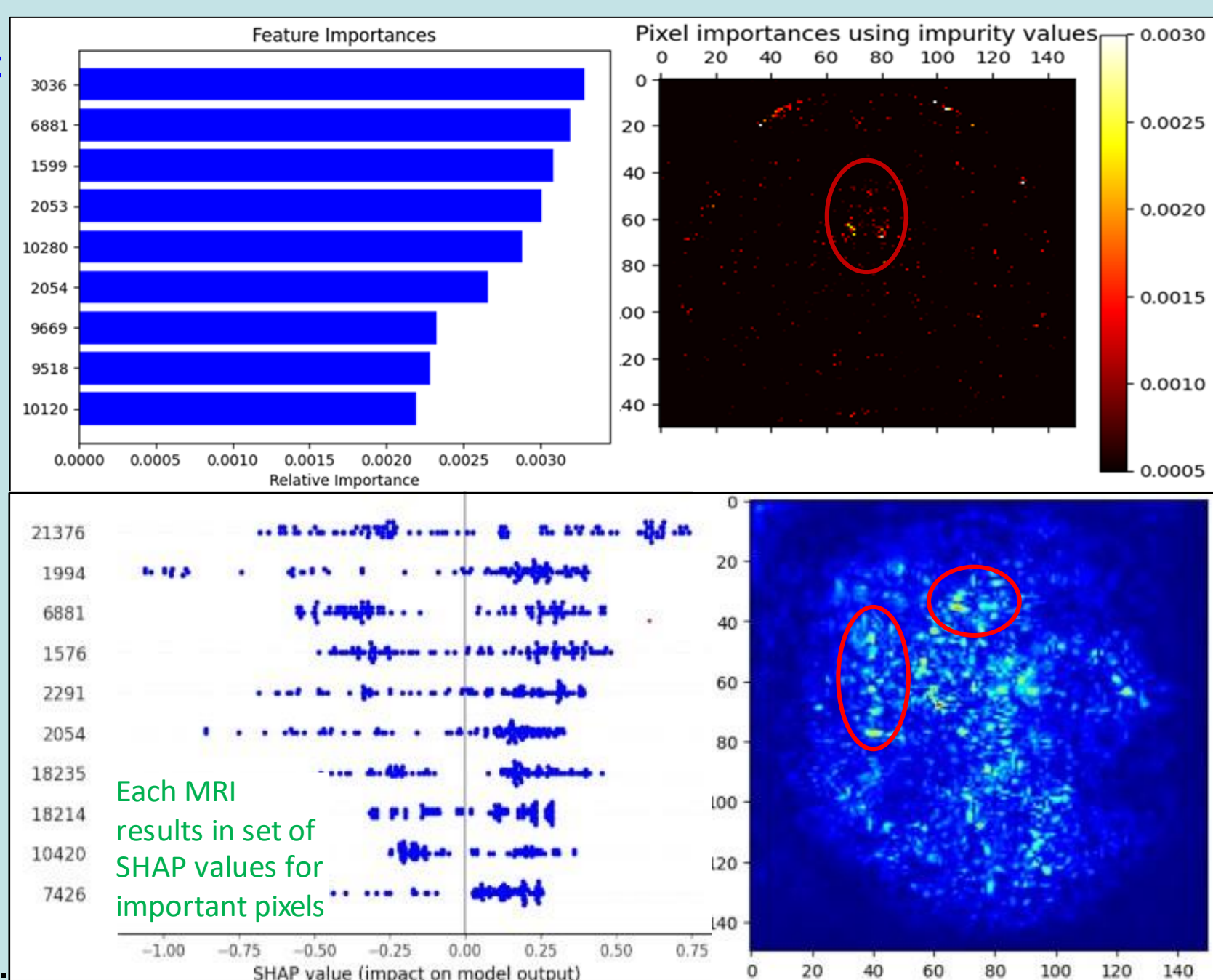
Patient Level

SHAP Dependence:

Relationship between a feature and the model's prediction.

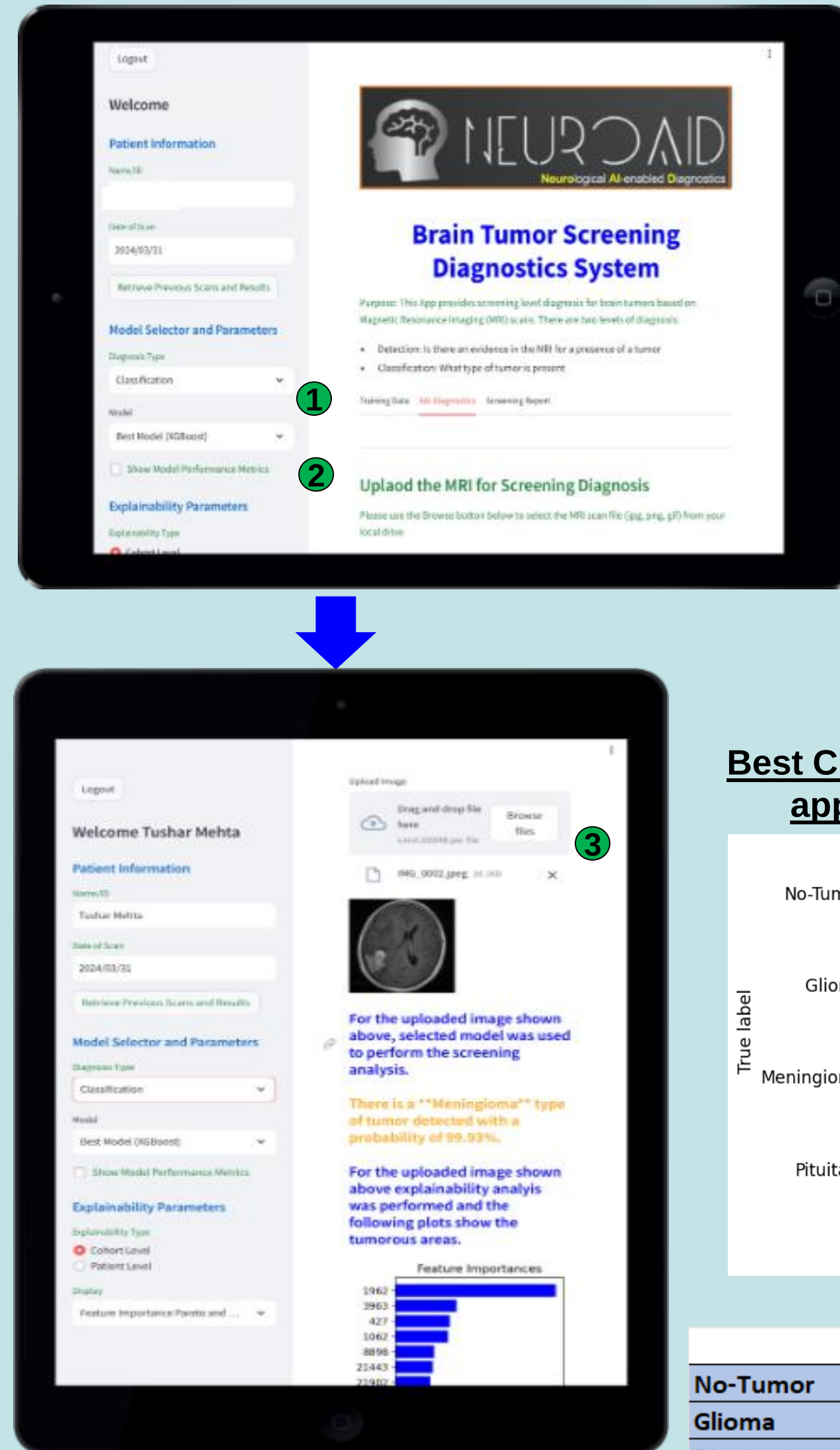
Saliency Maps:

High intensity pixels represent the regions that correlate to the corresponding targets.



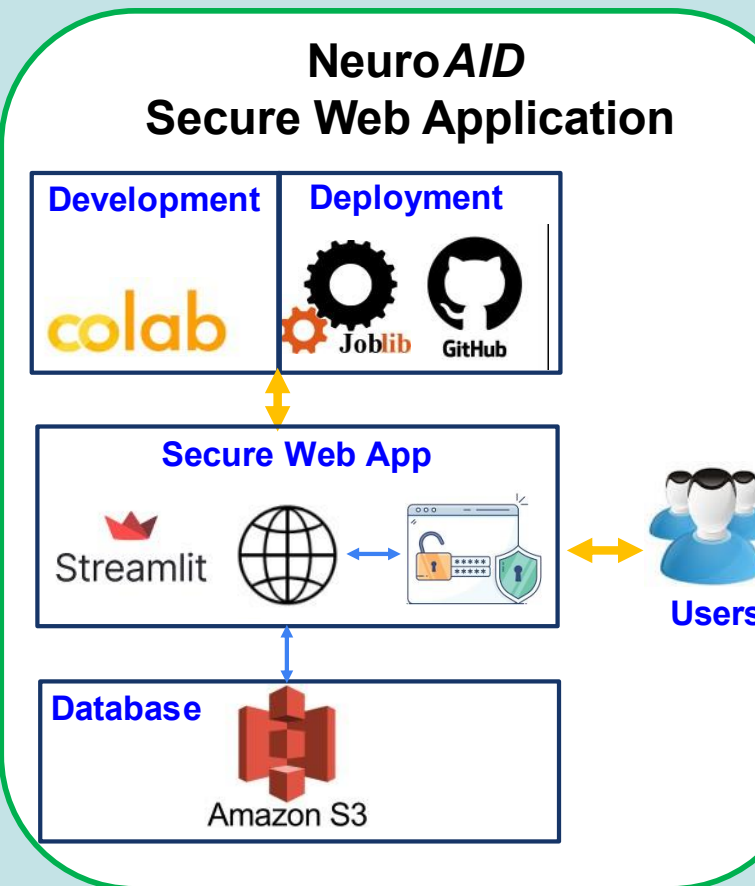
CLINICAL APPLICATION

3-step Screening Diagnosis



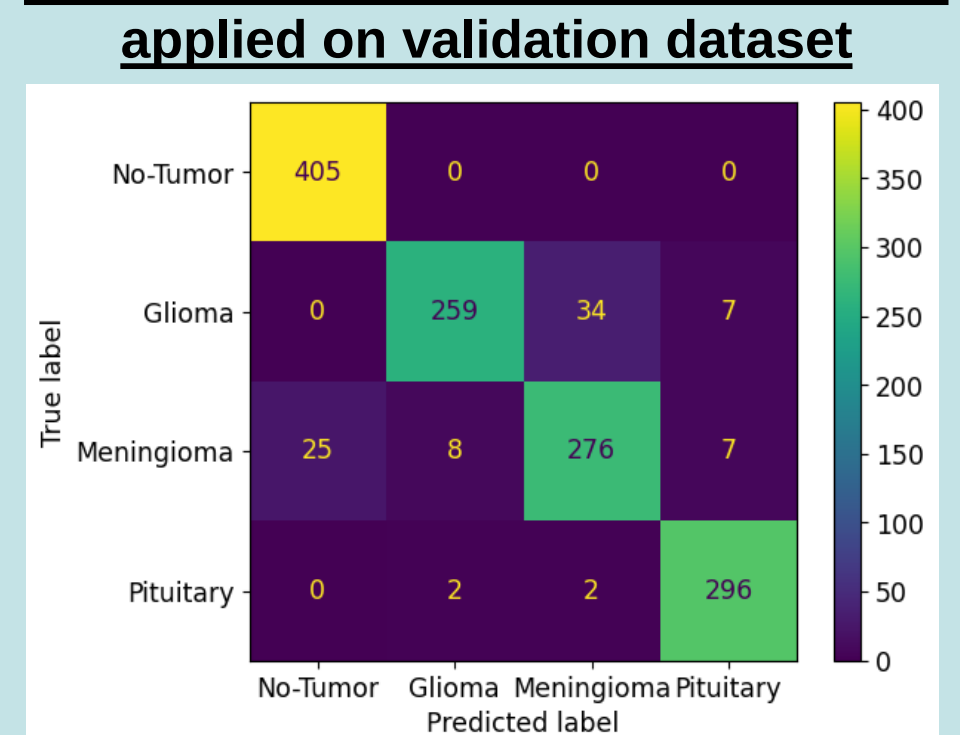
Outputs: (a) Probability for the presence and type of tumor (b) Medical insights (explainability)

System Architecture



Field Trial

Best Classification Model (XGBoost) applied on validation dataset



Classification Report

	Precision	Recall	F1-Score	Support
No-Tumor	0.94	1	0.97	405
Glioma	0.96	0.86	0.91	300
Meningioma	0.88	0.87	0.88	316
Pituitary	0.95	0.99	0.97	300
Overall accuracy			0.94	1321
macro avg	0.94	0.93	0.93	1321
weighted avg	0.94	0.94	0.93	1321

CONCLUSIONS

Data supported original hypothesis that transfer-learning based hybrid models can be trained on MRI scans to classify brain tumors into classes: no tumors, glioma, meningioma & pituitary.

KEY RESULTS

- Proposed novel hybrid model** using features from ResNet50 model as inputs to k-neighbor classifier outperforms other ML and TL algorithms with an accuracy of 98% (better than standard MRI diagnostics).
- XGBoost with learning rate of 0.41 was found to be the best classification model in terms of achieving the highest overall accuracy of 0.94 and high F1-scores for all three tumor types.
- Feature-importance and saliency maps provide differentiated neurological insights as leading indicators for tumors, highlighting the affected areas, not available from visual diagnostics.
- Best-in-class predictive models were implemented in a secure NeuroAID (Neurological AI-enabled Diagnostics) system, which democratizes MRI analysis by making it accessible to everyone across the globe.

CLINICAL APPLICATIONS

- Digital Radiologist:** Intelligent and explainable capabilities in NeuroAID provide confidence and equitable access for patients at point-of-care in under-served rural areas and under-developed countries as well as instantaneous diagnosis in emergency rooms.
- Targeted Therapeutics:** Time-based analysis of MRI scans can be used to track tumor growth and guide the personalized treatment.

ONGOING AND FUTURE RESEARCH

Ideas for improvement and reducing classification error:

- Continue field trials with protected patient data & include those in training.
- Hybrid model: Optimal selection of fused features from TL models.
- Next Steps:**
 - Hybrid models for the size and grade of the tumors based on the areas and location of high intensity pixels.
 - Screening for other neurological disorders: Model training to include other disorders, providing comprehensive diagnostics in a universal system.
 - Explore the idea of integrating NeuroAID capabilities into MRI systems.